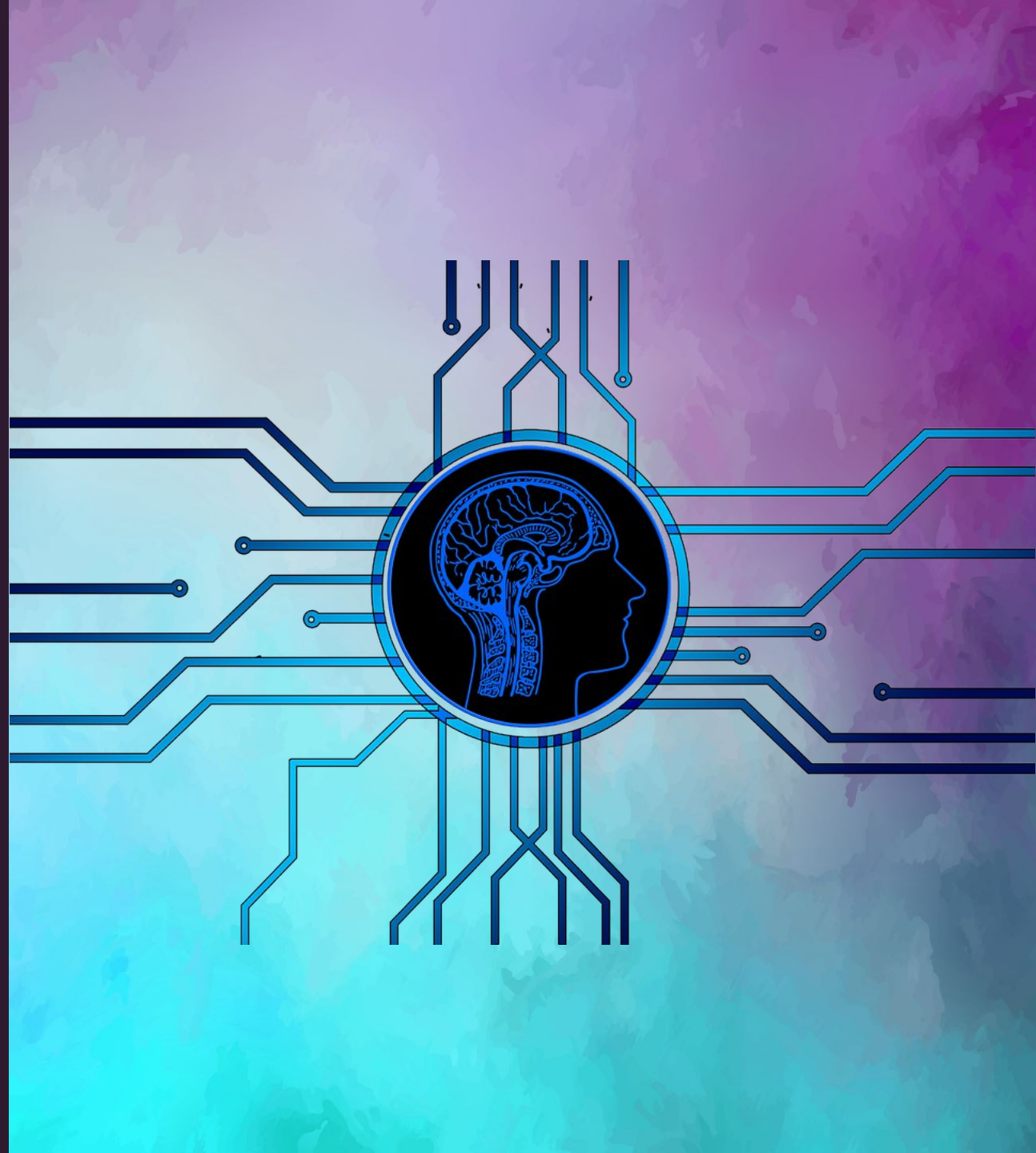


Human-Centered Artificial Intelligence + Modern AI

SIT112 | Intro to Data Science & AI

Lecture Week 11

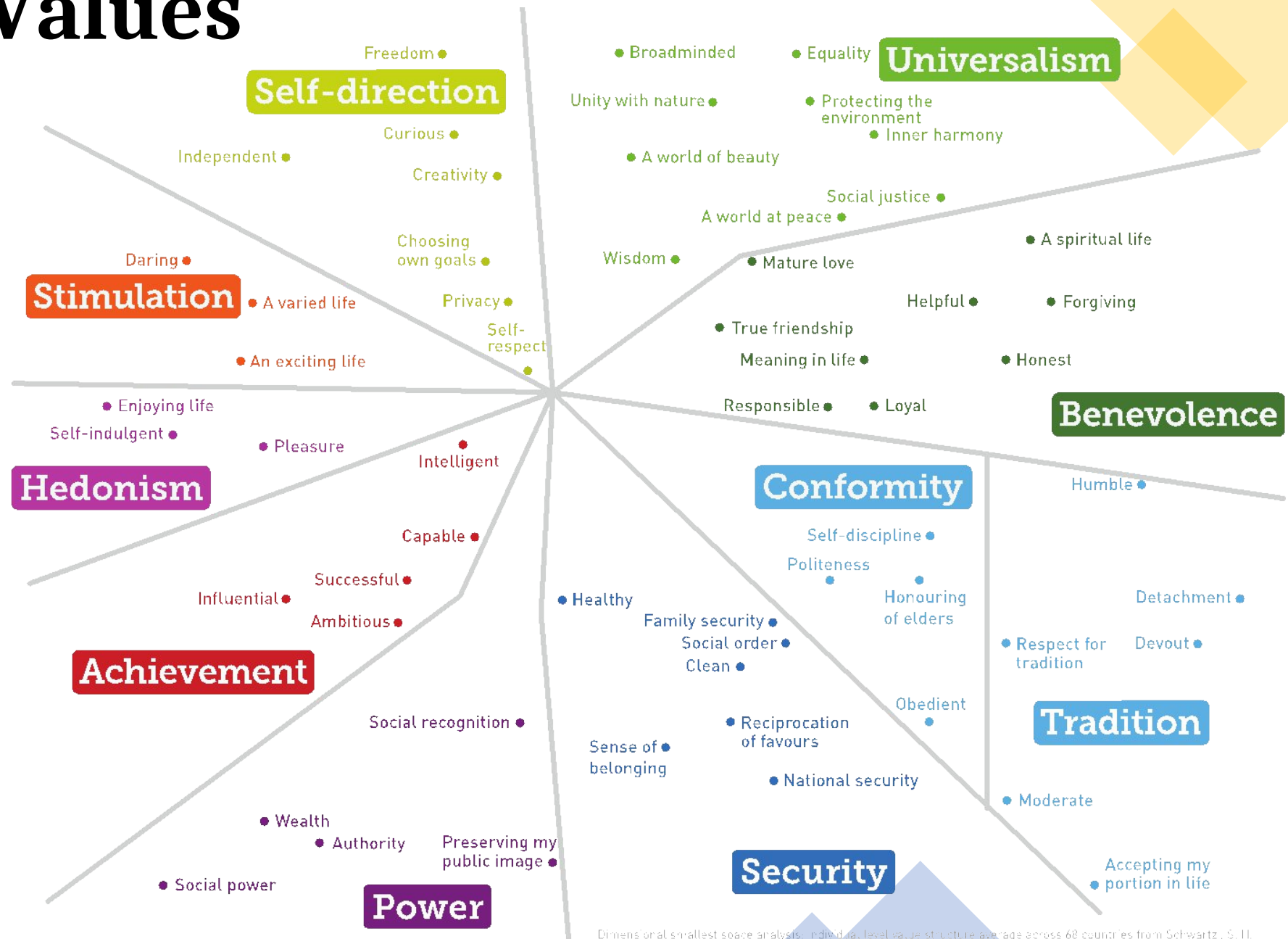
School of Information Technology, Deakin University



Objectives

-  **Explain Human-Centred AI**
-  **Recognise ethical and social harm**
-  **Describe responsible AI principles**
-  **Define an AI agent**
-  **Distinguish an AI model from an AI agent**
-  **Design a responsible AI agent**

Human Values



Dimensional smallest space analysis: individual level value structure average across 68 countries from Schwartz, S. H. (2006). Basic human values: Theory, measurement, and applications. *Annuaire français de sociologie*, 42, 249-288.



What could go wrong?

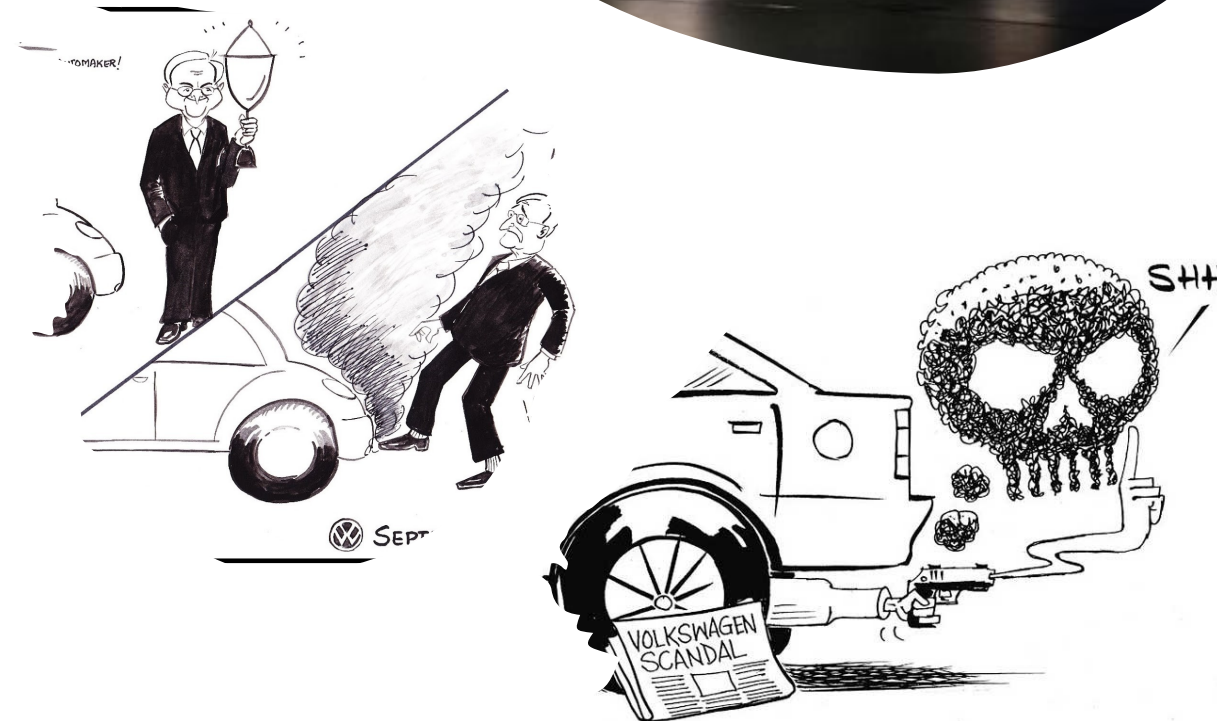


Hurricane Irma Price Gouging

- 5 hours flight from Miami to Phoenix
- Prices skyrocketed from \$547 to over \$3,258 USD
- New York Times: “*there are no ethics valves built into the system that prevent an airline from overcharging*”

Volkswagen Fuel Emission Fiasco - 2015

- A **deceptively** designed **software** suppressed the fuel emission control system on the road while keeping it engaged in the lab.
- Breaching the company's value "Responsible Thinking"
- 25% drop in sales in a year.
- Resignation of the CEO





Ministers

Department of Health and Aged Care

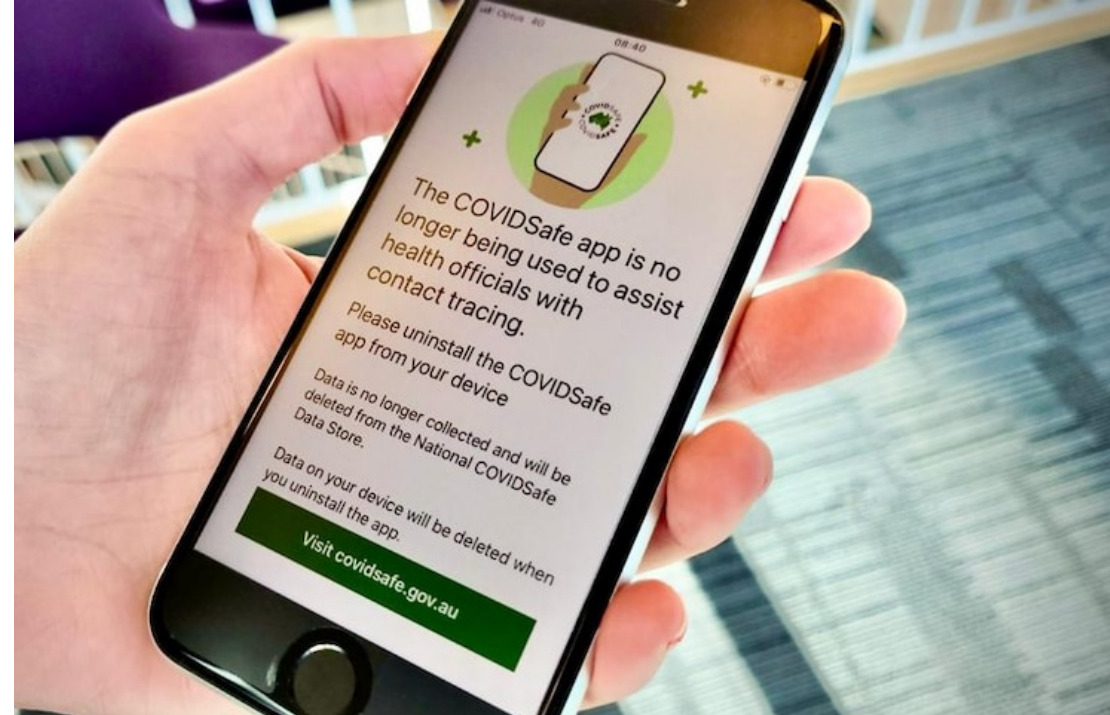
[Home](#)[Media centre](#)[Mark Butler](#)[Anika Wells](#)[Ged Kearney](#)[Emma McBride](#)[Mal](#)

[Ministers](#) > [The Hon Mark Butler MP](#) > [Minister Butler's media](#)

Failed COVIDSafe app deleted

The Albanese Government has acted to delete the wasteful and ineffective COVIDSafe app.

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LIVE BLOG

Chinese ambassador to Australia, Xiao Qian, is giving a speech at the National Press Club

The COVIDSafe app is dead – but was it ever really alive?

• Preakness 2024 Israel-Hamas war Alice Stewart dies at 58 Sean 'Diddy' Combs Usyk beats Fury

EVEN WHEN THE NEWS IS FREE, JOURNALISM IS NOT.
SUPPORT INDEPENDENT, FACT-BASED JOURNALISM.

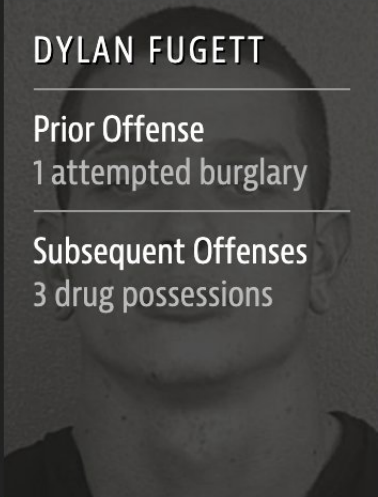
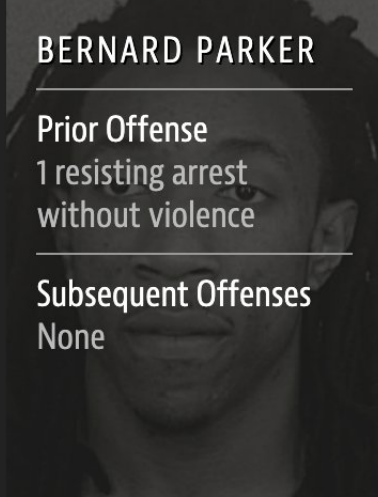
DONATE

Meta, TikTok and other social media CEOs testify in heated Senate hearing on child exploitation



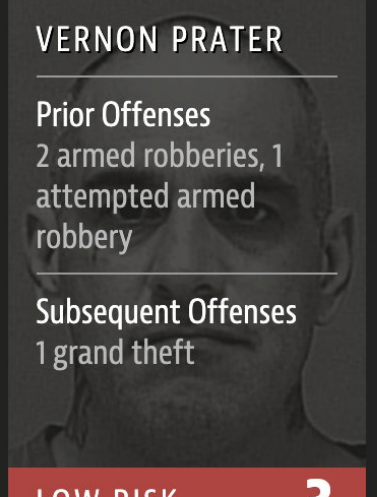
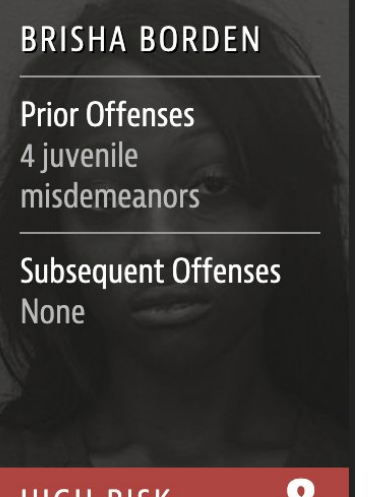
Racial bias in criminal-risk assessment

Two Drug Possession Arrests

 DYLAN FUGETT Prior Offense 1 attempted burglary Subsequent Offenses 3 drug possessions LOW RISK 3	 BERNARD PARKER Prior Offense 1 resisting arrest without violence Subsequent Offenses None HIGH RISK 10
---	---

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

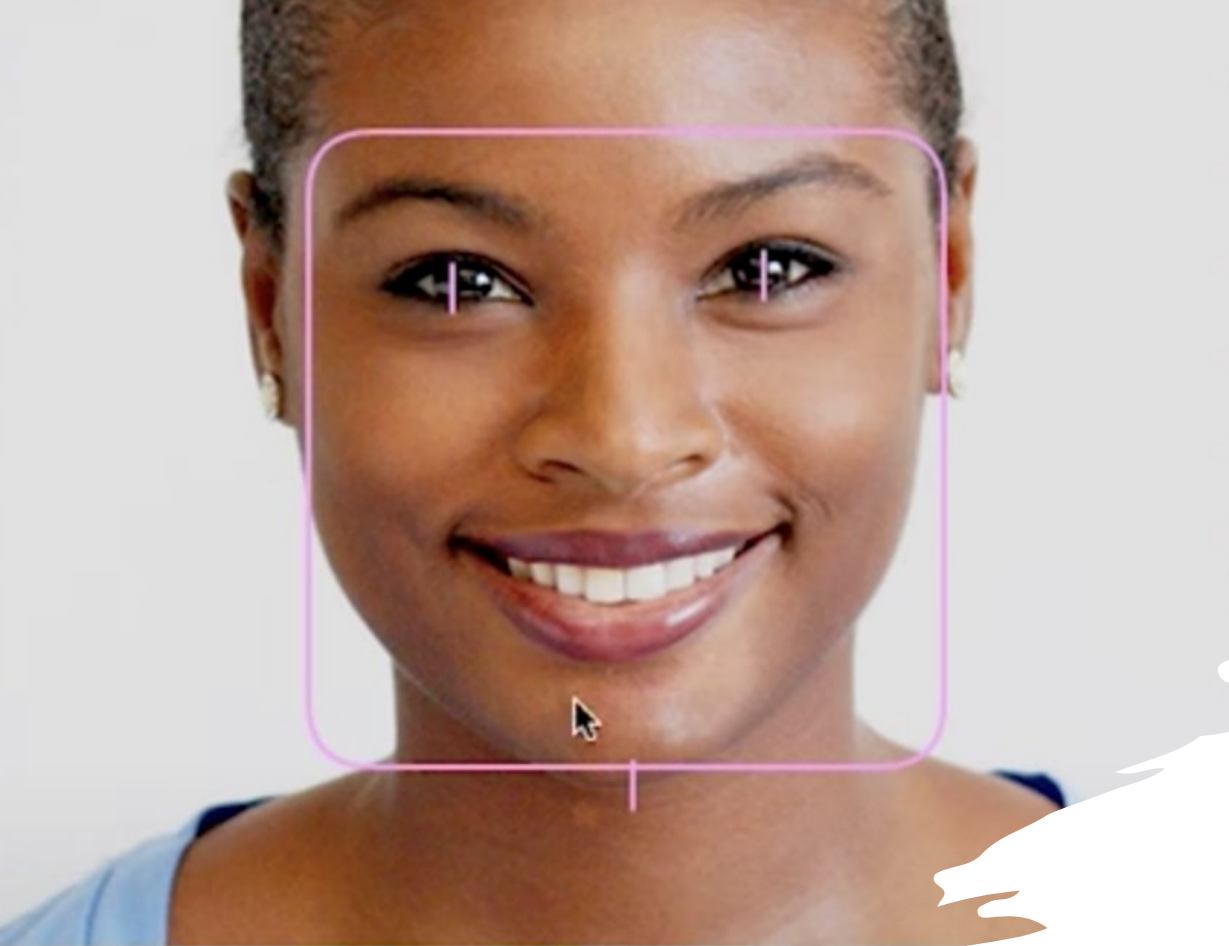
Two Petty Theft Arrests

 VERNON PRATER Prior Offenses 2 armed robberies, 1 attempted armed robbery Subsequent Offenses 1 grand theft LOW RISK 3	 BRISHA BORDEN Prior Offenses 4 juvenile misdemeanors Subsequent Offenses None HIGH RISK 8
--	---

Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

Falsely labeling black defendants as potential criminals by the **recidivism assessment models of the US criminal justice system**.

<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>



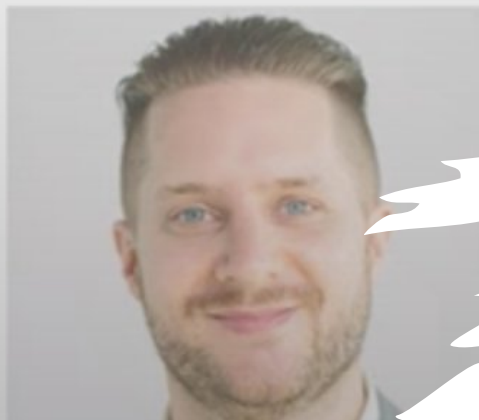
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Higher error rate in detecting dark-skinned women compared to light-skinned men in facial recognition software.

Coded Gaze

<https://www.youtube.com/watch?v=162VzSzzoPs>

https://www.youtube.com/watch?time_continue=299&v=TWWsW1w-BVo&feature=emb_logo



Three simple stick figures are drawn in a row. The first figure on the left is blue and has a triangular body, representing a female. The middle figure is red and has a rectangular body, representing a male. The third figure on the right is purple and is shaped like a question mark, representing a gender that is unknown or being questioned.

☐ **Male** ☐ **Female** ☒ *Other*

Human-Centered AI: What is it?



The remarkable progress in AI has opened the doors to new opportunities, and some dark possibilities.



However, a bright future awaits AI researchers, developers, business leaders, policy-makers, and others who build on AI algorithms by including HCAI strategies of design and testing.



This enlarged vision can shape the future of technology so as to better serve human values and needs; the goal is not to replace people but to empower them by making design choices that give humans control over technology.

Human-Centered AI Framework: Reliable, Safe, and Trustworthy Systems

Attributes that make **HCAI** systems reliable, safe, and trustworthy.



Reliable systems based on sound software



Safety culture through business management



Trustworthy certification by independent

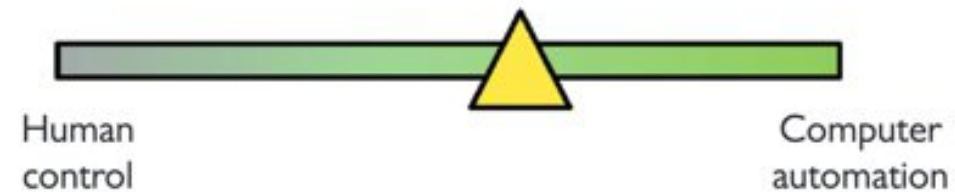


Regulation by government

Human-Centered AI Framework: Rising above the Levels of Automation

- Automated systems can be wonderful, but strategies to ensure **activation when needed** and **prevent inadvertent activations** contribute to reliable, safe, and trustworthy operation that increases user acceptance.
- **High levels of human control** and **high levels of automation** increase **reliability, safety, and trustworthiness** in applications.
- Achieving these goals, especially for **complex, poorly understood problems**, will dramatically **increase human performance** while supporting **human self-efficacy, creativity, responsibility, and social connections**.

Level	Description
5 (High)	Full autonomy: equal to that of a human driver, in every driving scenario.
4	High automation: fully autonomous vehicles perform all safety-critical driving functions in certain areas and under defined weather conditions.
3	Conditional automation: driver shifts "safety critical functions" to the vehicle under certain traffic or environmental conditions.
2	Partial automation: at least one driver assistance system is automated. Driver is disengaged from physically operating the vehicle (hands off the steering wheel AND foot off the pedal at the same time).
1	Driver assistance: most functions are still controlled by the driver, but a specific function (like steering or accelerating) can be done automatically by the car.
0 (Low)	No automation: human driver controls all: steering, brakes, throttle,



Misleading one-dimensional thinking suggests that designers must choose between human control and computer automation.

Human-Centered AI Framework: Two-Dimensional HCAI Framework

- The **desired** goal is often to create **designs that are in the upper right quadrant**.
- The **lower right quadrant** is home to relatively **mature, well-understood systems for predictable tasks**, e.g., car automatic transmission.
- For **poorly understood and complex** tasks with varying contexts of use, the **upper right quadrant** is needed.

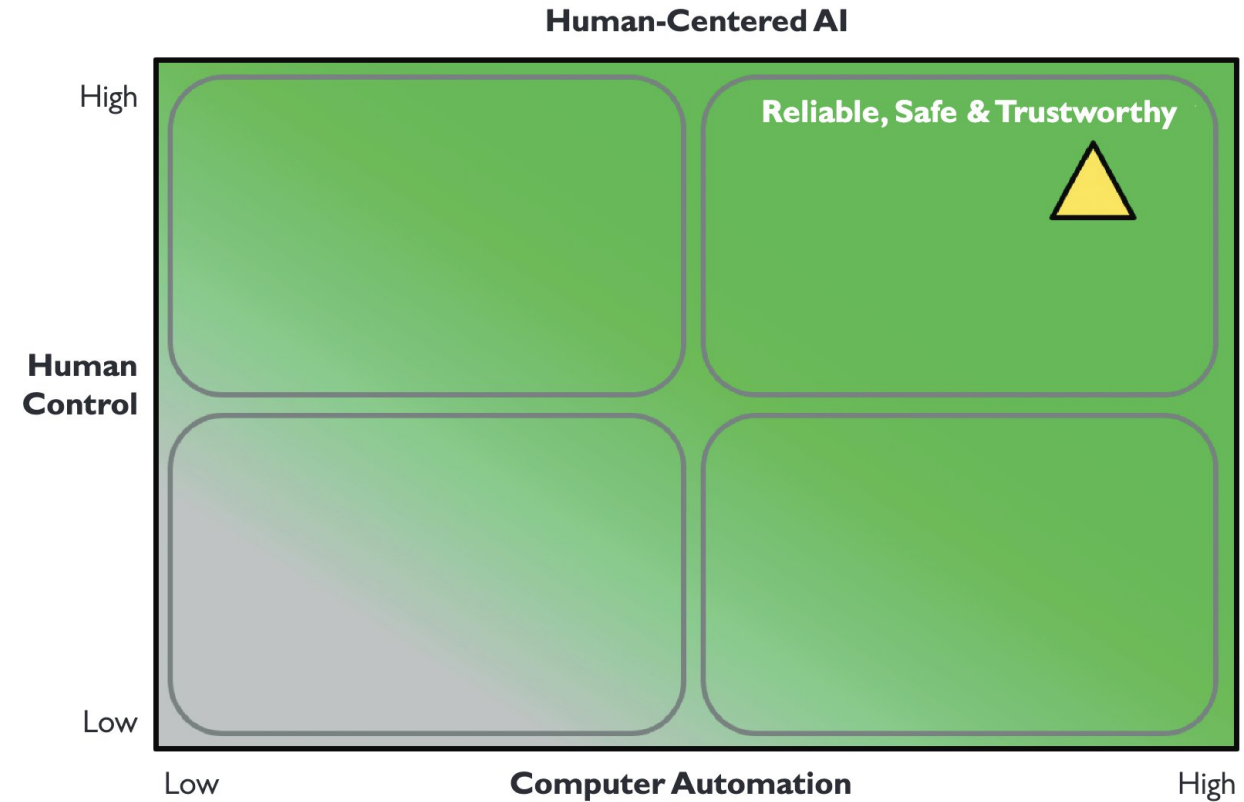


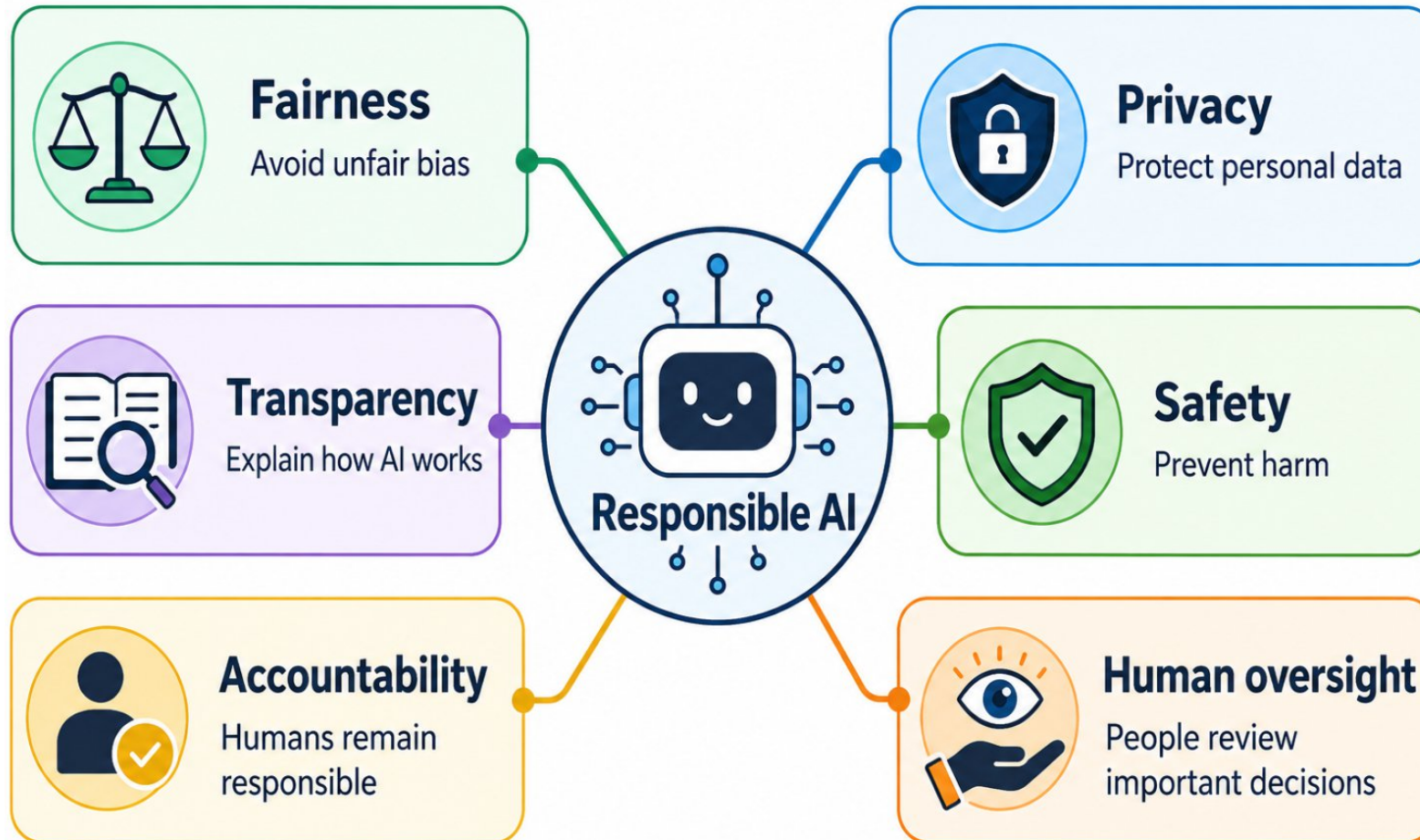
Fig 8.2 Two-dimensional framework with the goal of reliable, safe, and trustworthy designs, which are achieved by a high level of human control and a high level of computer automation (yellow triangle).

Human-Centered AI: Strategies for Ethical AI

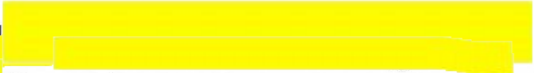
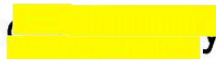
- **Inclusive Design**: Engaging **diverse** groups in AI development to reflect a wide range of human values and perspectives.
- **Ethical AI Governance**: Developing and enforcing **guidelines, policies, and regulations** to guide ethical AI development and use.
- **Continuous Monitoring**: Regularly **assessing** AI systems for ethical compliance and societal impact.



Part 2: Responsible AI

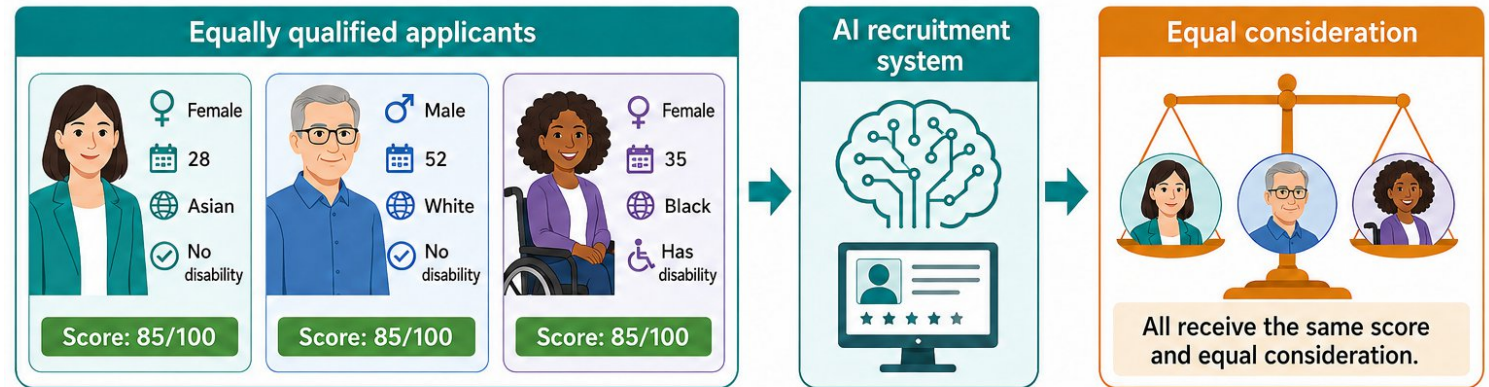


What Is Responsible AI?

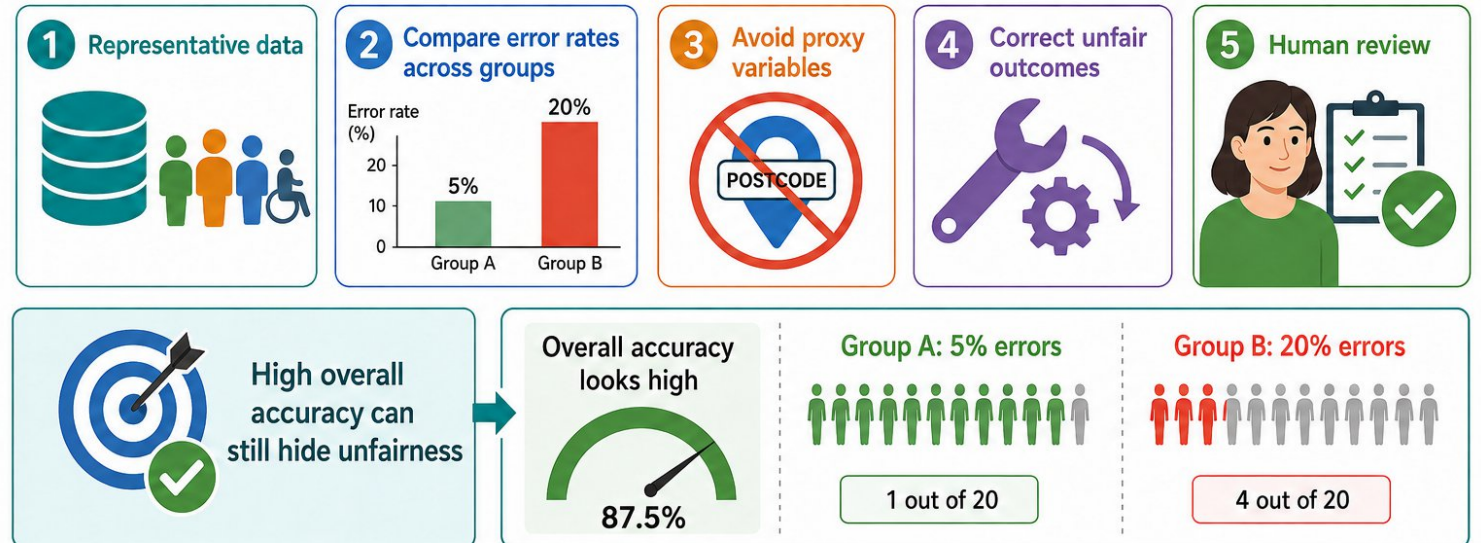
Principles of Responsible AI		
	Fairness	AI should not unfairly  against individuals or groups.
	Transparency	People should  and  its role.
	Privacy	Personal information must be collected and protected responsibly.
	Safety and reliability	AI should work  and avoid causing  .
	Accountability	 must remain responsible for its outcomes.
	Human oversight	People should  question, and  important AI decisions.

Fairness in AI Recruitment

AI systems should treat **people** and groups **equitably** and should not create or reinforce **unfair discrimination**.



Five ways to improve fairness:



AI system can have high overall accuracy but still be unfair if it makes more mistakes for one population group than another.

Privacy and Data Protection

Although related, **privacy** and **data protection** are not exactly the same.

Privacy: respecting people's right to control their **personal information**. It concerns **whether** information should be **collected or used in the first place**.

Data protection: keeping that information secure and using it **only** for an appropriate and **agreed-upon purpose**.

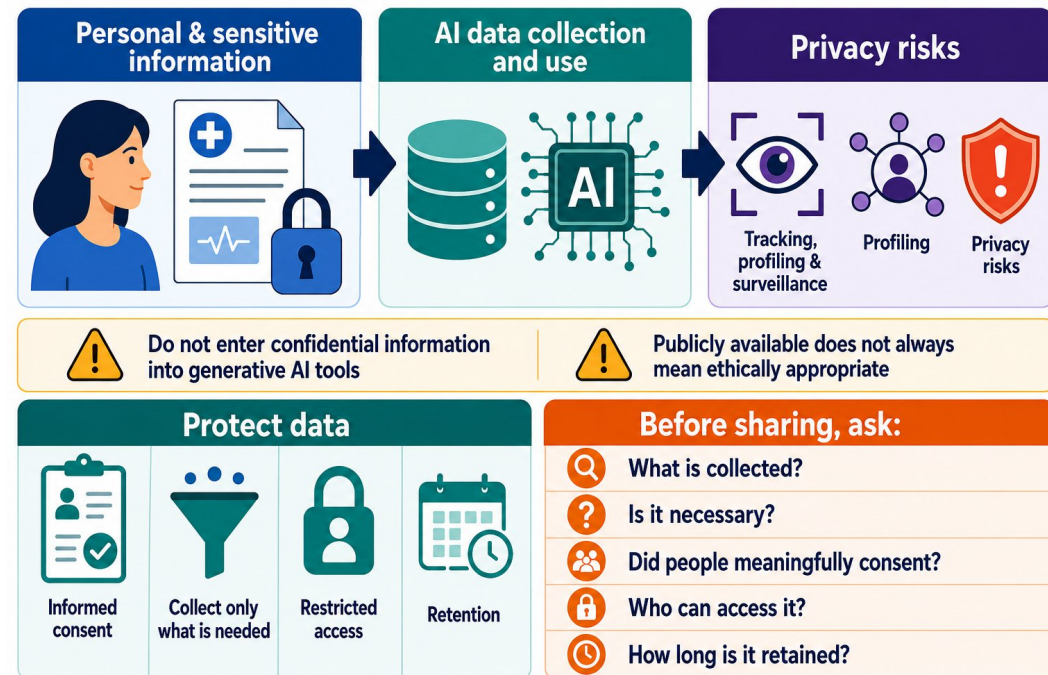
For example,

privacy asks:

Should the university collect this information about a student?

Data protection asks:

How will the university store it securely and prevent unauthorised access?



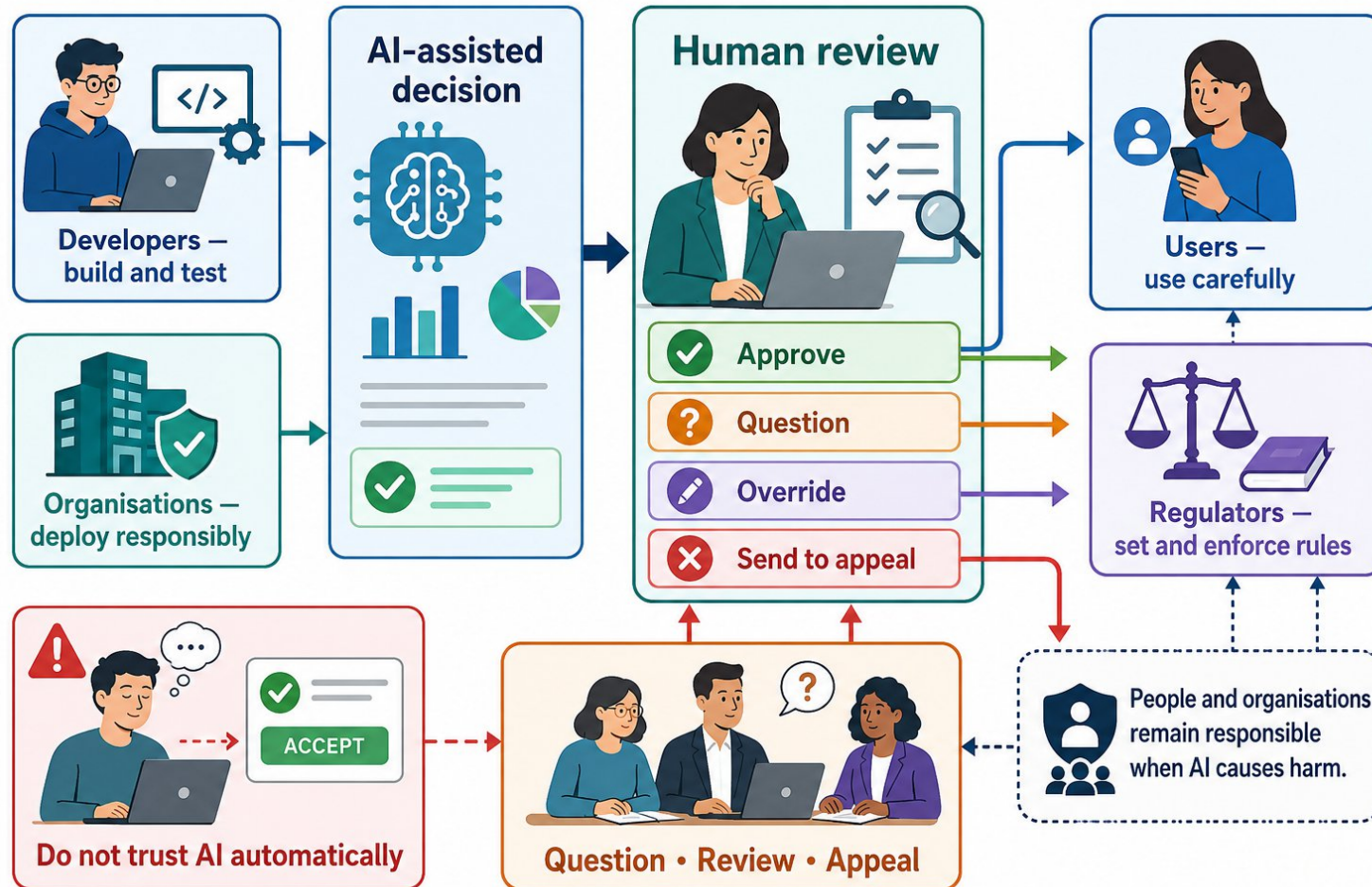
Safety, Reliability and Misuse

AI system **can be technically reliable but still unsafe** or misused.

For example, a **deepfake generator** could work extremely **reliably**; it **produces highly realistic videos** every time, but that same reliability could make its **misuse more harmful**.

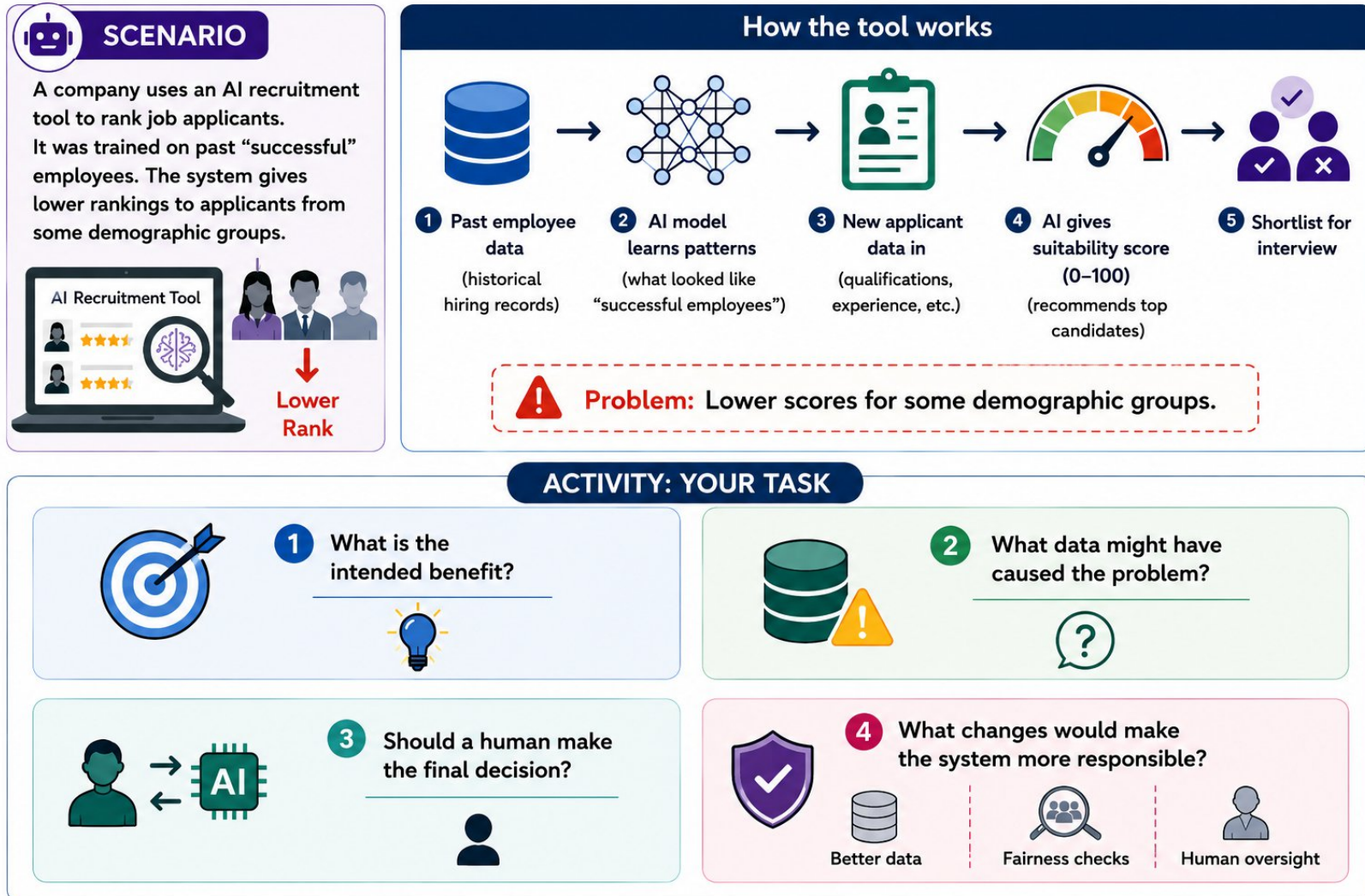


Accountability and Human Oversight

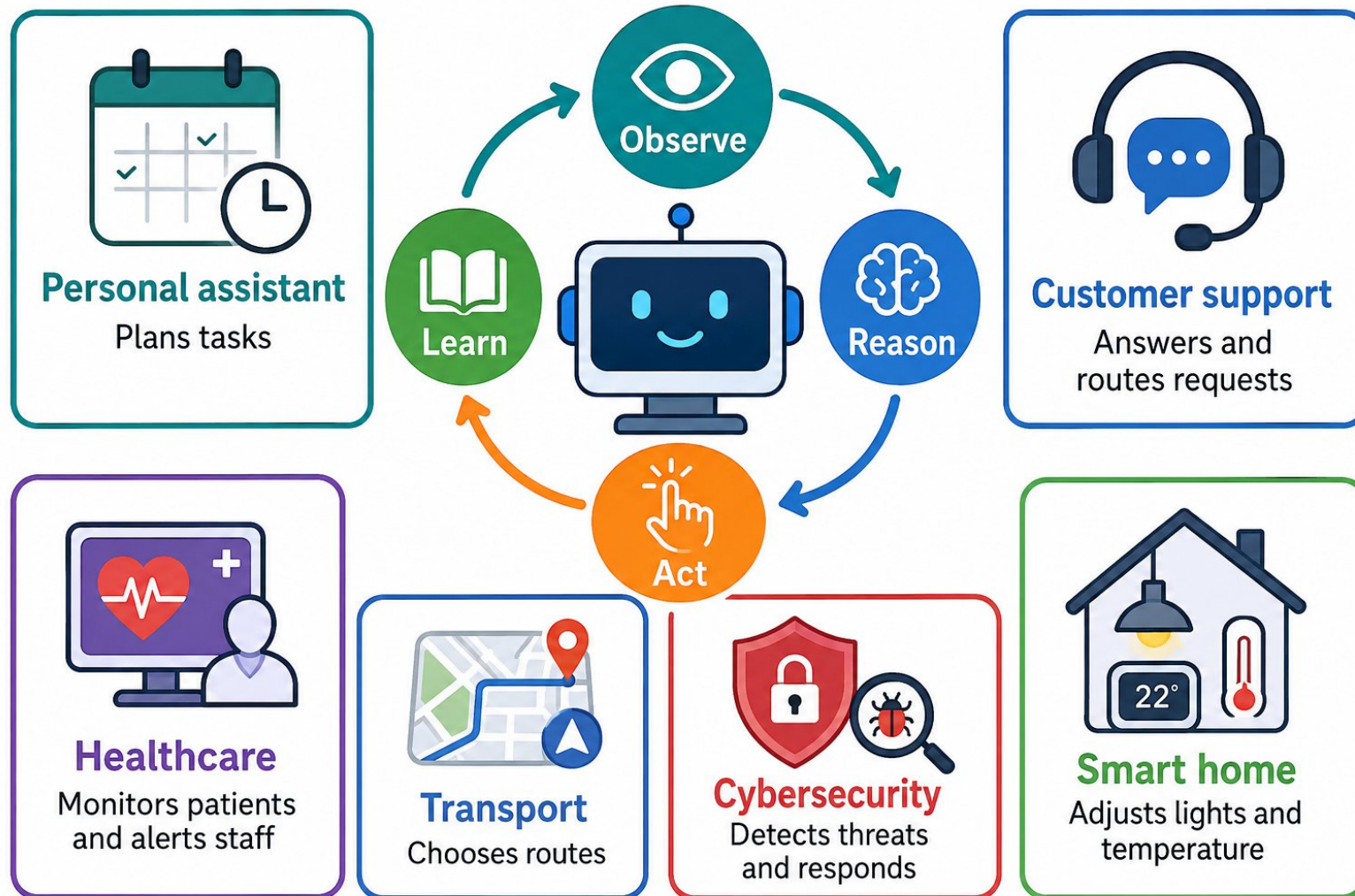


Activity:

Would you trust this AI recruiter?



Part 3: AI agents and real-life use cases



What Is an Agent?

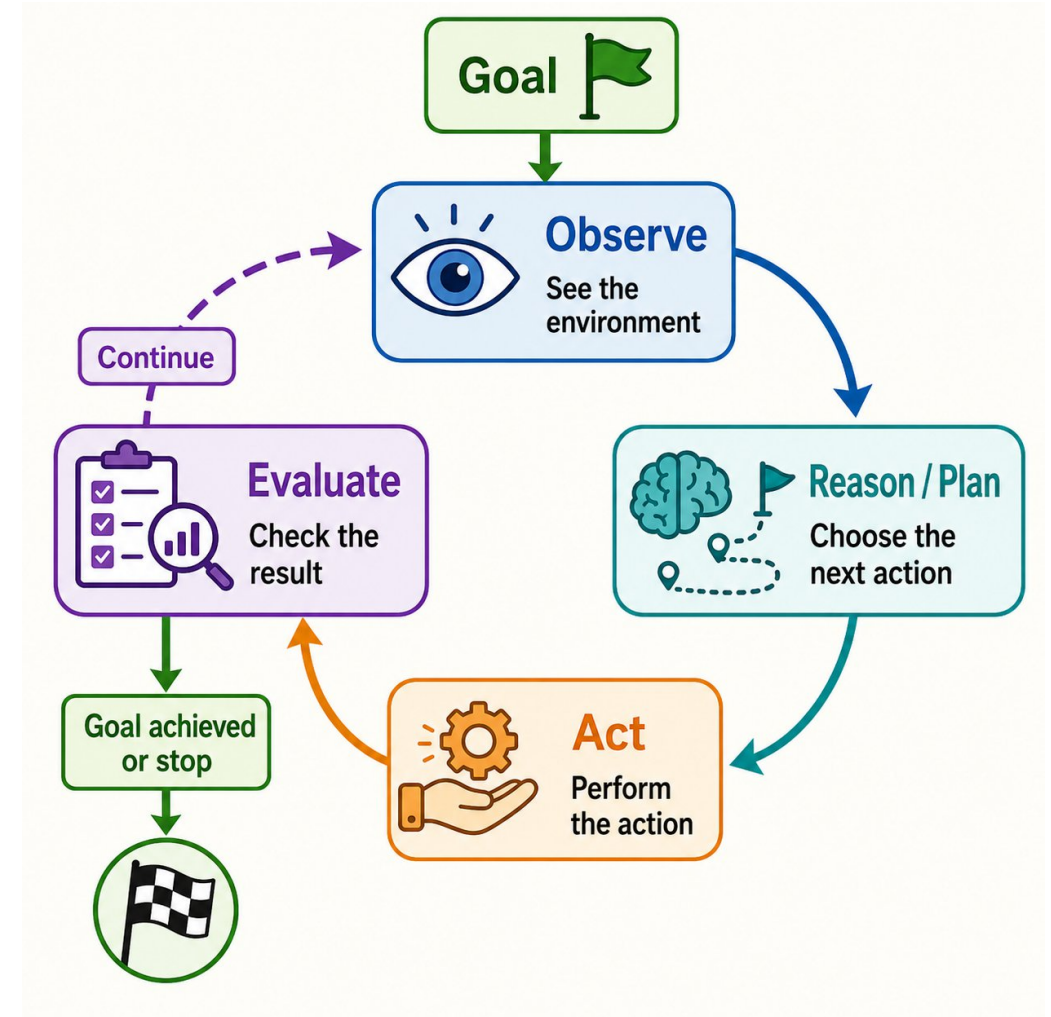
An **AI agent** is a system that works toward a goal by observing its environment, planning and taking actions, and evaluating the results.

- It repeats this cycle.
- It stops when the (i) goal is achieved, a (ii) stopping condition is reached or (iii) human assistance is required.

Observe → Reason → Act → Learn

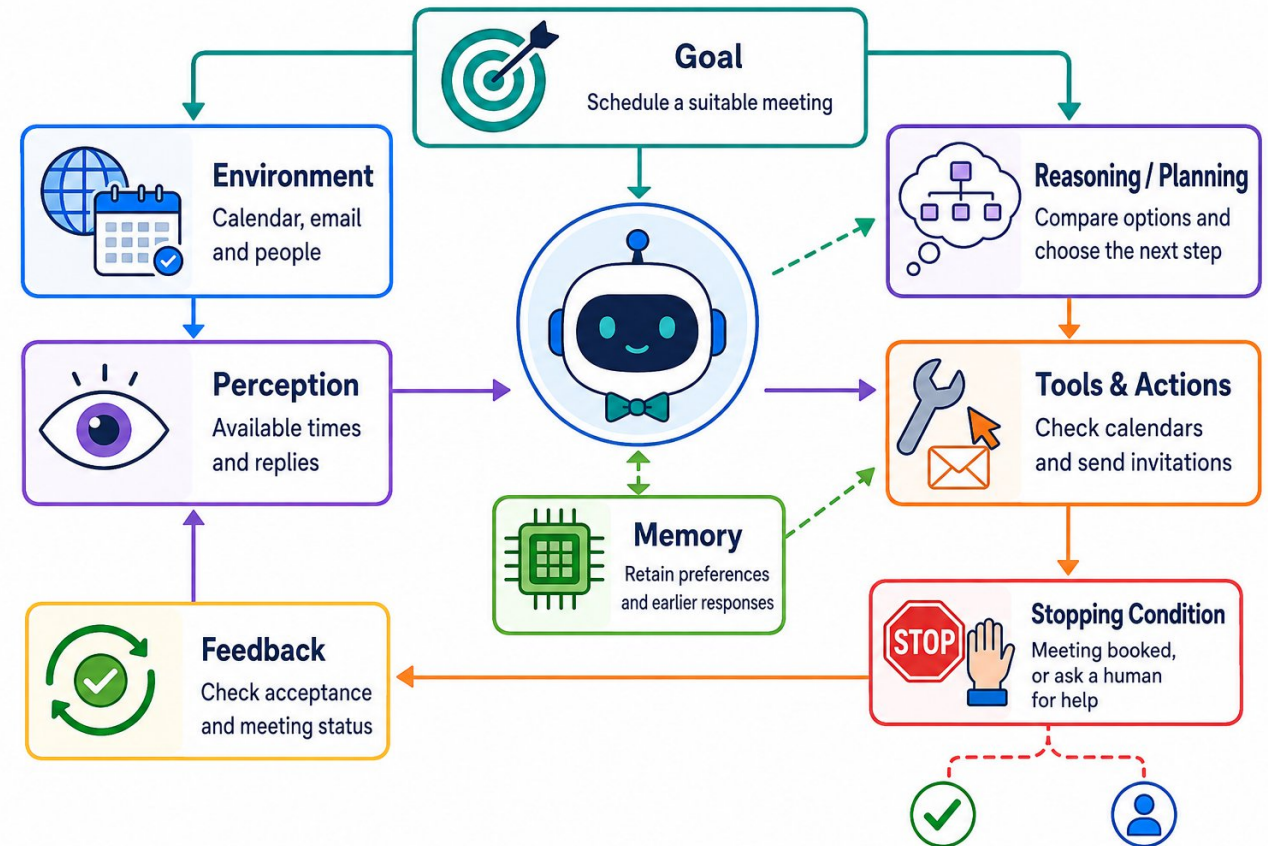
The Agent Cycle

- **Goal**
The agent begins with a **defined objective**, for example, scheduling a suitable meeting.
- **Observe**
It **collects information** about the environment, such as calendar availability, emails or user instructions.
- **Reason or plan**
It compares **possible actions** and **selects the most appropriate** next step.
- **Act**
It **performs an action**, for example, checking calendars or sending an invitation.
- **Evaluate**
It **checks** whether the action produced the desired result.



Components of an AI Agent

- **Goal:** Defines what the agent must achieve.
- **Environment:** The **setting** in which the agent operates.
- **Perception:** The **information** the agent can **observe** or receive.
- **Reasoning/Planning:** How it compares options and selects its next step.
- **Tools and Actions:** What the agent is **permitted and able to do**.
- **Memory:** Information retained from **earlier interactions** or actions.
- **Feedback:** **Evidence** used to check whether its actions are working.
- **Stopping Condition:** Determines when the agent should **finish or request human help**.



Example of an Agent

Task: Arrange a meeting with my research team next week.



AI model vs. AI agent

 AI model	 AI agent
 Usually responds to an input	 Works towards a goal
 Produces an output	 Can perform one or more actions
 Often completes one interaction	 May operate through multiple steps
 Does not necessarily use external tools	 May use tools, databases, APIs or software
 Example: summarising an email	 Example: reading an email, checking a calendar and proposing a meeting

Risks Specific to Agents

AI agents introduce greater risks because they can use tools and **take actions** that **affect people, data**, and systems.

 	Incorrect or unauthorised action	 	Ambiguous goal misinterpreted
 	Excessive personal-data access	 	Unreliable tools or information
 	Continues after an error	 	Manipulated by malicious instructions
 	Small mistakes compound into serious harm	 	Acts without sufficient human approval

References

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Perera, Harsha, et al. "A study on the prevalence of human values in software engineering publications, 2015--2018." *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering*. 2020.



End of lecture ...